Chapter 3

TECHNIQUES OF INTEGRATION

(Page 261-330, Calculus Volume 2)



MAE101

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3.1 Integration by Parts

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3.7 Improper Integrals

TECHNIQUES OF INTEGRATION

3.1 Integration by Parts

In this section, we will learn:

How to integrate complex functions by parts.

INTEGRATION BY PARTS

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\,dx$$

- Let $u = f(x)$ and $v = g(x)$.
- Then, the differentials are:

$$du = f'(x)\,dx \quad \text{and} \quad dv = g'(x)\,dx$$

Thus, by the Substitution Rule, the formula for integration by parts becomes:

$$\int u\,dv = uv - \int v\,du$$

INTEGRATION BY PARTS

Evaluating both sides of Formula 1 between a and b, assuming f' and g' are continuous, and using the FTC, we obtain:

$$\int_a^b f(x)g'(x)\,dx = f(x)g(x)\Big|_a^b - \int_a^b g(x)f'(x)\,dx$$

INTEGRATION BY PARTS

Example 1

Find $I = \int x \sin x dx$

INTEGRATION BY PARTS

Example 1

Let $u = x \qquad dv = \sin x\,dx$

Then, $du = dx \qquad v = -\cos x$

Using Formula 2, we have:

$$\int x \sin x\,dx = \int \overset{u}{x} \overbrace{\sin x\,dx}^{dv} = \overset{u}{x} \overbrace{(-\cos x)}^{v} - \int \overbrace{(-\cos x)}^{v} \overset{du}{dx}$$

$$= -x\cos x + \int \cos x\,dx$$

$$= -x\cos x + \sin x + C$$

INTEGRATION BY PARTS

Example 2

Evaluate $I = \int e^x \sin x \, dx$

- e^x does not become simpler when differentiated.
- Neither does sin x become simpler.

INTEGRATION BY PARTS

Example 2

Nevertheless, we try choosing

$$u = e^x \text{ and } dv = \sin x$$

- Then, $du = e^x\,dx$ and $v = -\cos x$.

INTEGRATION BY PARTS

Example 2

So, integration by parts gives:

$$\int e^x \sin x\,dx = -e^x \cos x + \int e^x \cos x\,dx$$

INTEGRATION BY PARTS

Example 2

The integral we have obtained, $\int e^x \cos x \, dx$,
is no simpler than the original one.

- At least, it's no more difficult.

- Having had success in the preceding example integrating by parts twice, we do it again.

INTEGRATION BY PARTS

Example 2

This time, we use

$$u = e^x \text{ and } dv = \cos x\, dx$$

Then, $du = e^x\, dx$, $v = \sin x$, and

$$\int e^x \cos x\, dx = e^x \sin x - \int e^x \sin x\, dx$$

INTEGRATION BY PARTS

Example 2

we get:

$$\int e^x \sin x\,dx = -e^x \cos x + e^x \sin x - \int e^x \sin x\,dx$$

- This can be regarded as an equation to be solved for the unknown integral.

INTEGRATION BY PARTS

Example 2

Adding to both sides $\int e^x \sin x \, dx$,
we obtain:

$$2\int e^x \sin x \, dx = -e^x \cos x + e^x \sin x$$

INTEGRATION BY PARTS

Example 2

Dividing by 2 and adding the constant of integration, we get:

$$\int e^x \sin x \, dx = \frac{1}{2} e^x (\sin x - \cos x) + C$$

Example 3

Suppose f(x) is continuous and differentiable, f(1)=4 and $\int_0^1 f(x)dx = 5$ Find $\int_0^1 xf'(x)dx$

a	4/5
b	5/4
c	1
d	None of the others
e	-1

Example 4

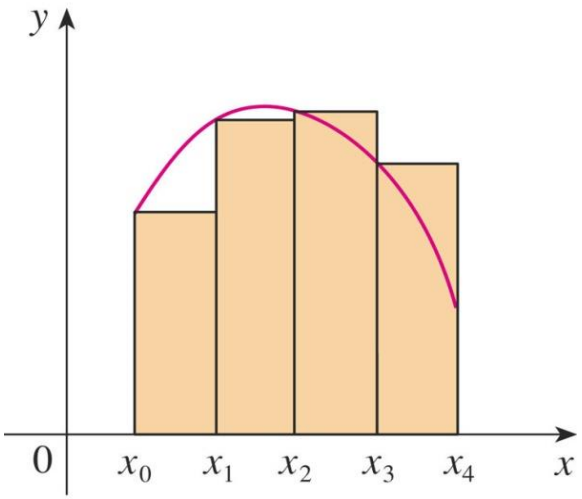
Suppose f(x) is continuous and differentiable, f(1)=3, f(3)=1 and $\int_1^3 xf'(x)dx = 13$

What is the average value of f on the interval [1,3]?

3.6

Numerical Integration

Left endpoint Method

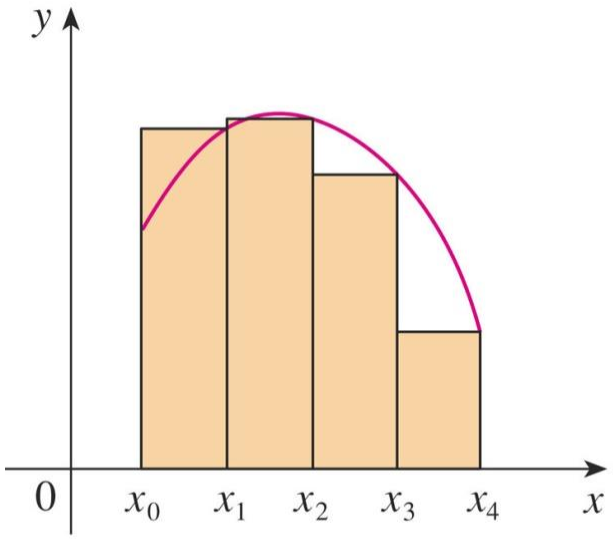


(a) Left endpoint approximation

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$$\int_a^b f(x)dx \approx \Delta x[\, f(x_0) + f(x_1) + \ldots + f(x_{n-1})]$$

Right endpoint Method

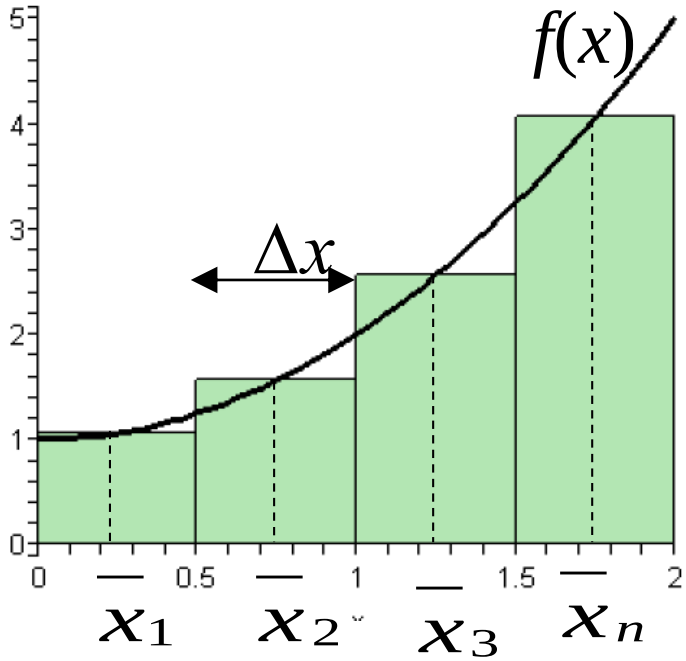


(b) Right endpoint approximation

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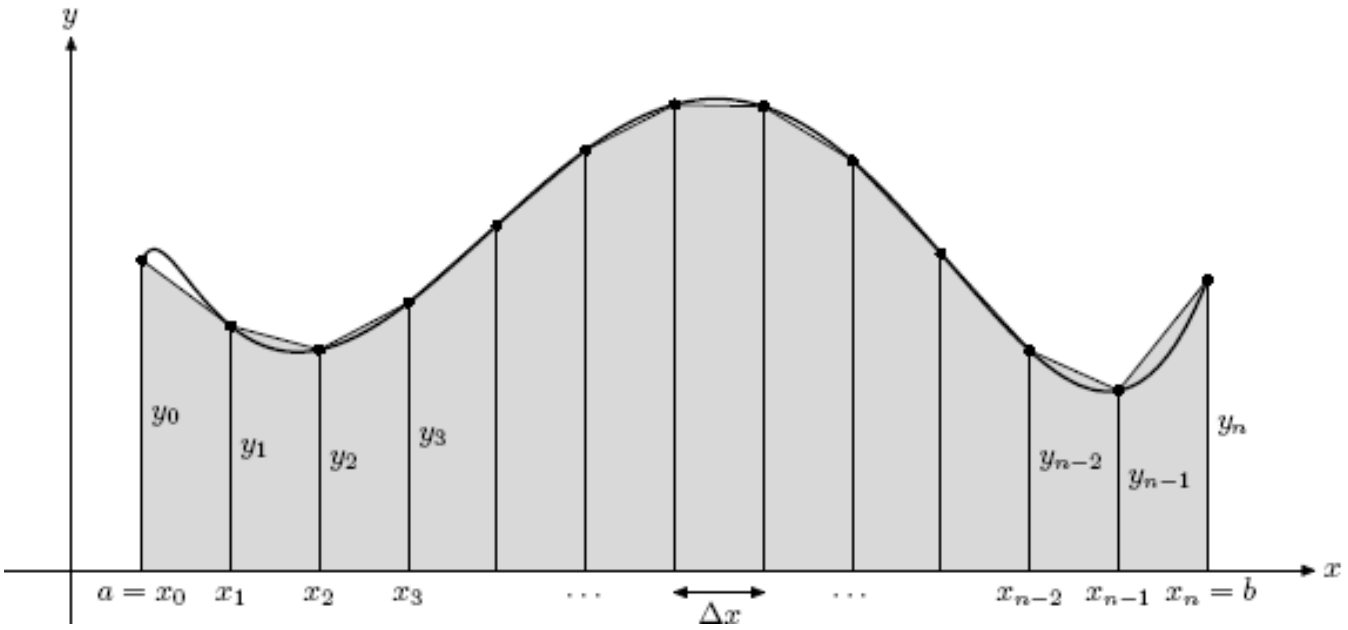
$$\int_a^b f(x)dx \approx \Delta x[\, f(x_1) + f(x_2) + \ldots + f(x_n)]$$

Midpoint Method



$$\int_a^b f(x)dx \approx \Delta x[\, f(\bar{x}_1) + f(\bar{x}_2) + \ldots + f(\bar{x}_n)]$$

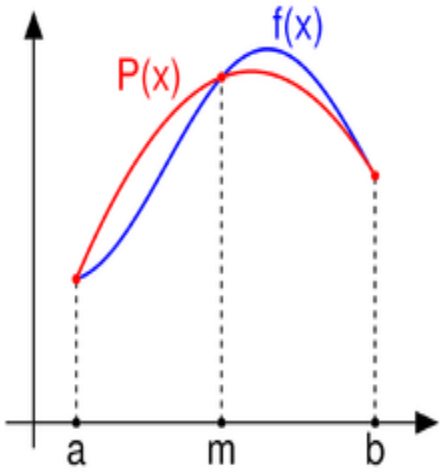
Trapezoidal Method

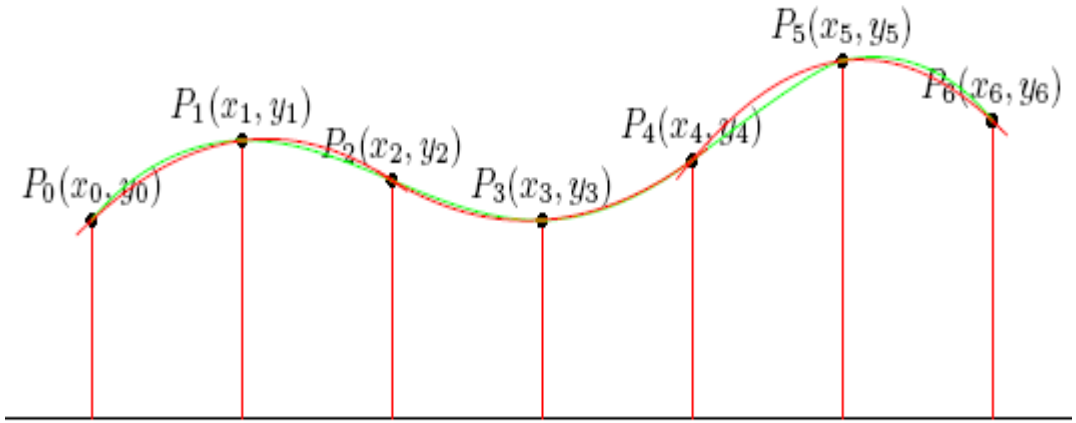


$$\int_a^b f(x)dx \approx \frac{\Delta x}{2}[f(x_0)+f(x_1)]+\ldots+\frac{\Delta x}{2}[f(x_{n-1})+f(x_n)]$$

$$\approx \frac{\Delta x}{2}[f(x_0)+2f(x_1)+\ldots+2f(x_{n-1})+f(x_n)]$$

Simpson Method





$$\int_a^b f(x)dx \approx \frac{\Delta x}{3}[f(x_0)+4f(x_1)+f(x_2)]+\ldots+\frac{\Delta x}{3}[f(x_{n-2})+4f(x_{n-1})+f(x_n)]$$

$$\approx \frac{\Delta x}{3}[f(x_0)+4f(x_1)+2f(x_2)+\ldots+4f(x_{n-1})+f(x_n)]$$

Example

Approximate the integral $\int_1^2 (1/x)\,dx$ with $n = 8$, using:

a. Left/Right endpoints

b. Midpoints

c. Trapezoidal method

d. Simpson method

Estimate error for Midpoint and Trapezoidal method

- Suppose $|f''(x)| \le K$ for $a \le x \le b$.
- If E_T and E_M are the errors in the Trapezoidal and Midpoint Rules, then

$$|E_T| \le \frac{K(b-a)^3}{12n^2} \quad \text{and} \quad |E_M| \le \frac{K(b-a)^3}{24n^2}$$

Estimate error for Simpson method

- Suppose $| f^{(4)}(x) | \le K$ for $a \le x \le b$.
- If E_S is the error in the Simpson method, then

$$|E_s| £ \frac{K(b - a)^5}{180n^4}$$

Example

How large should we take *n* in order to guarantee that the Trapezoidal, Midpoint Rule, Simpson rule approximations for

$$\int_1^2 (1/x)\,dx$$

are accurate to within 0.0001?

$|f''(x)| \le 2$ for $1 \le x \le 2$

Accuracy to within 0.0001 means that error < 0.0001

Trapezoidal: Choose smallest n so that: $\frac{2(1)^3}{12n^2} < 0.0001$

$\rightarrow n = 41$

Midpoint: $\frac{2(1)^3}{24n^2} < 0.0001 \rightarrow n = 30$

Simpson: $\left|f^{(4)}(x)\right| = \left|\frac{24}{x^5}\right| £ 24$

$\rightarrow \frac{24(1)^5}{180n^4} < 0.0001 \rightarrow n = 8$

TECHNIQUES OF INTEGRATION

3.7 Improper Integrals

In this section, we will learn:

How to solve definite integrals

where the interval is infinite and

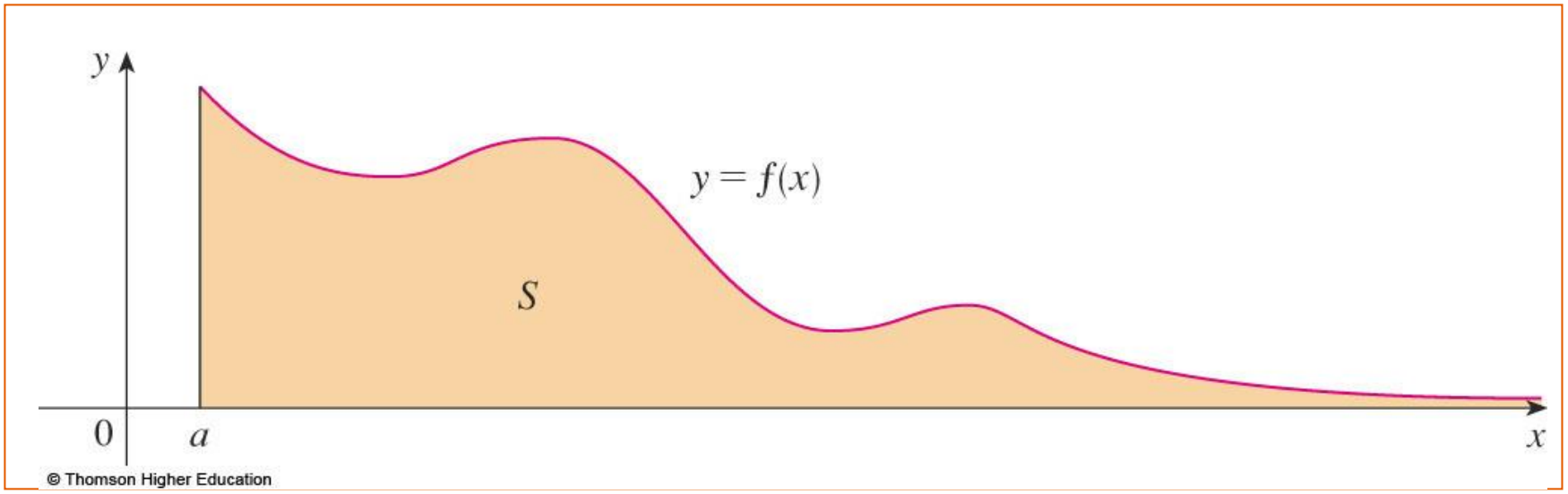
where the function has an infinite discontinuity.

Definition

If $\int_a^t f(x)\,dx$ exists for every number $t \geq a$, then

$$\int_a^{\infty} f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx$$

provided this limit exists (as a finite number).



Definition

If $\int_t^b f(x)\,dx$ exists for every number $t \le a$, then

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{t \to -\infty} \int_t^b f(x)\,dx$$

provided this limit exists (as a **finite number**).

CONVERGENT AND DIVERGENT

Definition

The improper integrals $\int_a^{+\infty} f(x)\,dx$ and $\int_{-\infty}^{b} f(x)\,dx$ are called:

- Convergent if the corresponding limit exists.
- Divergent if the limit does not exist.

Definition

If both $\int_a^{+\infty} f(x)\,dx$ and $\int_{-\infty}^{a} f(x)\,dx$ are convergent,

then we define:

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^{a} f(x)\,dx + \int_a^{+\infty} f(x)\,dx$$

- Here, any real number a can be used.

Example 1

For what values of p is the integral $\int_1^{\infty} \frac{1}{x^p}\,dx$ convergent?

- Convergent if $p > 1$
- Divergent if $p \leq 1$

Example 2

Investigate the convergence of the improper integrals:

(a) $\int_{-\infty}^{0} x^2 e^{x^3} dx$

(b) $\int_{-\infty}^{\infty} x^2 e^{x^3} dx$

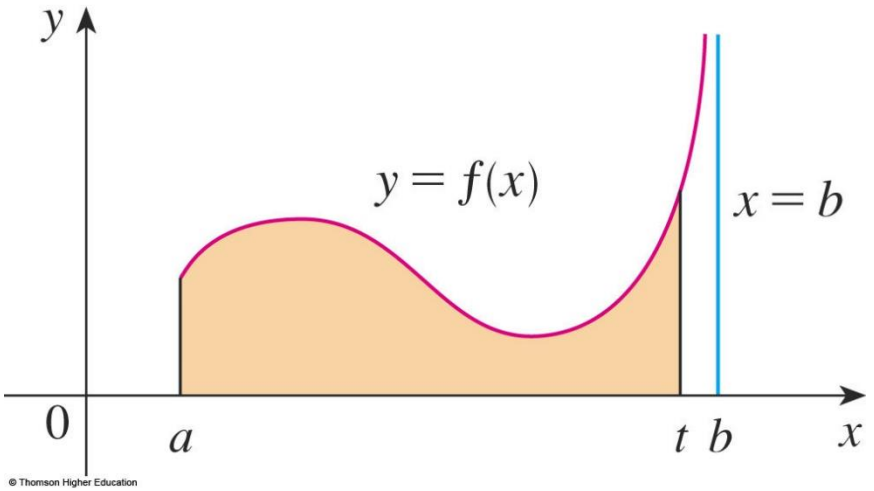
IMPROPER INTEGRAL OF TYPE 2

Definition

If f is continuous on $[a, b)$ and is discontinuous at b, then

$$\int_a^b f(x)\,dx = \lim_{t \to b^-} \int_a^t f(x)\,dx$$

if this limit exists (as a finite number).



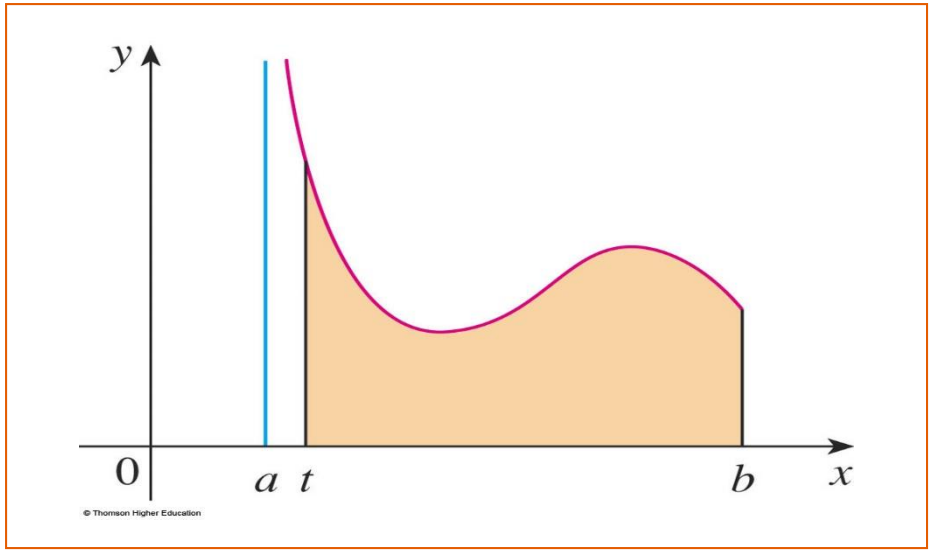
IMPROPER INTEGRAL OF TYPE 2

Definition

If f is continuous on $(a, b]$ and is discontinuous at a, then

$$\int_a^b f(x)\,dx = \lim_{t \to a^+} \int_t^b f(x)\,dx$$

if this limit exists (as a finite number).



IMPROPER INTEGRAL OF TYPE 2

Definition

The improper integral $\int_a^b f(x)\,dx$ is called:

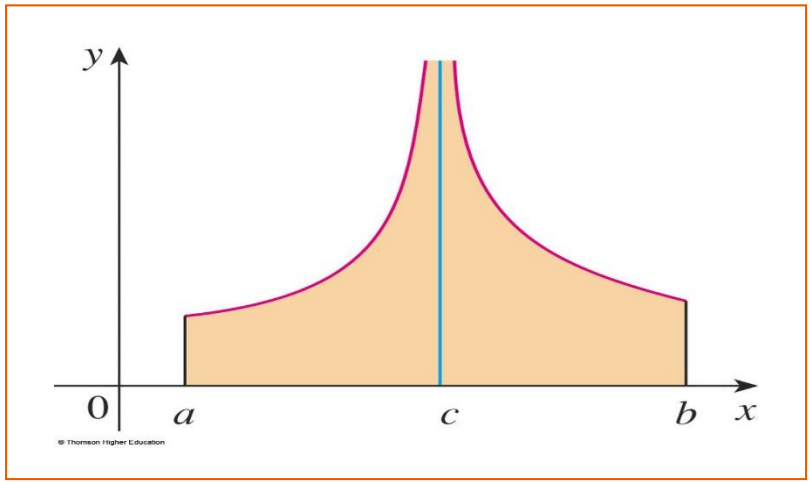
- Convergent if the corresponding limit exists.
- Divergent if the limit does not exist.

IMPROPER INTEGRAL OF TYPE 2

Definition

If f has a discontinuity at c, where $a < c < b$, and both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ are convergent, then we define:

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$



IMPROPER INTEGRAL OF TYPE 2

Example 1

Let b> 2. Investigate the convergence of the improper integrals:

$$\int_a^b \frac{dx}{(x-a)^p}$$

Answer: diverges if p $\geq$ 1 and converges if p<1.

IMPROPER INTEGRAL OF TYPE 2

Example 2

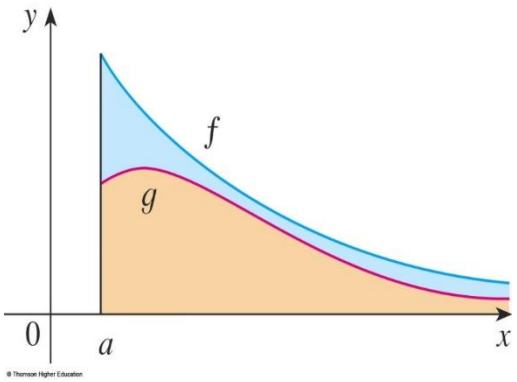
Investigate the convergence of the improper integral:

$$\int_0^3 \frac{dx}{x-1}$$

Suppose f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$.

a. If $\int_a^\infty f(x)\,dx$ is convergent, then $\int_a^\infty g(x)\,dx$ is convergent.

b. If $\int_a^\infty g(x)\,dx$ is divergent, then $\int_a^\infty f(x)\,dx$ is divergent.



Note: A similar theorem is true for Type II integrals

Example

Does $I = \int_1^{\infty} \frac{|\cos(x)|\,dx}{x^2}$ converge?

We have,

$$0 \le \frac{|\cos(x)|}{x^2} \le \frac{1}{x^2}$$

and $\int_1^{\infty} \frac{dx}{x^2}$ converges

$\rightarrow I$ converges.

Example

Investigate the convergence of the improper integrals

(a) $\int_1^{\infty} \frac{\sqrt{x+2}}{x}\,dx$

(b) $\int_1^{\infty} \frac{\sqrt{x+2}}{x^3}\,dx$

(c) $\int_0^{\infty} e^{-x^2}\,dx$